

$$\mathcal{F}(S) := \{f: S \rightarrow \mathbb{R}\}$$

$$A \subseteq S$$

$$I_A = \{f: S \rightarrow \mathbb{R} \mid f(a) = 0 \ \forall a \in A\}$$

is an ideal of $\mathcal{F}(S)$.

If A is singleton let's say $A = \{a\}$.

$$\text{Then } I_A = \{f: S \rightarrow \mathbb{R} \mid f(a) = 0\}.$$

Now, let $g, h \in \mathcal{F}(S)$ &

$$gh \in I_A \Rightarrow (gh)(a) = 0$$

$$\Rightarrow g(a)h(a) = 0$$

$$\Rightarrow g(a) = 0 \text{ or } h(a) = 0$$

$$\Rightarrow g \in I_A \text{ or } h \in I_A.$$

$\Rightarrow I_A$ is a prime ideal.

$$C[a, b] = \{f: [a, b] \rightarrow \mathbb{R} \mid f \text{ is cts}\}$$

$$C[a, b] \leq \mathcal{F}([a, b])$$

Existence of a Maximal Ideal:

$$\text{Let } \Sigma = \{I : I \subsetneq A\} ; A \neq \langle 0 \rangle$$

$$\langle 0 \rangle \in \Sigma \Rightarrow \Sigma \neq \varnothing. \quad \textcircled{\#}$$

Let $I_1 \subseteq I_2 \subseteq I_3 \subseteq \dots$ be a chain in

$$\bigcup_{n=1}^{\infty} I_n \text{ is an ideal}$$

$$\text{If } \bigcup_{n=1}^{\infty} I_n = A \Rightarrow 1 \in I_n \text{ for some } n \in \mathbb{N}.$$

$$\Rightarrow I_n = A \text{ as } I_n \in \Sigma$$

$$\Rightarrow \bigcup_{n=1}^{\infty} I_n \in \Sigma \text{ and this is an upper}$$

bound for the chain $\textcircled{\#}$.

Then by Zorn's Lemma, Σ has a maximal element say $\mathfrak{m}$. If $\mathfrak{m}$ is not a maximal element $\exists$ a proper ideal $J \subsetneq A$ and

$$\mathfrak{m} \subsetneq J \subsetneq A \text{ but } J \subsetneq A \Rightarrow J \in \Sigma$$

this contradicts $\mathfrak{m}$ is maximal element of Σ .

Th^m: If A is a non-zero commutative ring with identity, then A has a maximal ideal.

$\uparrow$
needed in finding upper bound

Cor: If A is a commutative ring with identity & I is a proper ideal of A , then $\exists$ a maximal ideal containing I .

Pf: Take $I \subsetneq A$ & consider A/I .

Now, A/I is CRI. $\Rightarrow A/I$ has a maximal ideal $\mathfrak{m}$. Then we have $\varphi: A \rightarrow A/I$
 $a \mapsto a+I$

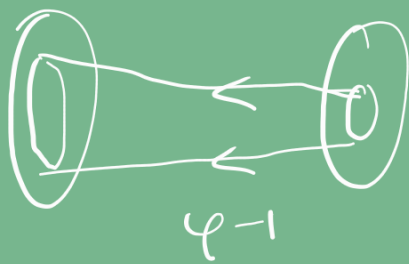
so, $\varphi^{-1}(\mathfrak{m}) \supseteq I$.

And as $\mathfrak{m} \neq A/I \Rightarrow \varphi^{-1}(\mathfrak{m}) \neq A$.

* $I \subsetneq \varphi^{-1}(\mathfrak{m}) \subsetneq A$ [$\because \mathfrak{m} \neq \{0\}$
 $\Rightarrow I \subsetneq \varphi^{-1}(\mathfrak{m})$]

Now, if $J \subsetneq A$ s.t.
 $I \subsetneq \varphi^{-1}(\mathfrak{m}) \subsetneq J \subsetneq A$.

$\mathfrak{m} \subsetneq J/I \subsetneq A/I$ *



$$\varphi(\varphi^{-1}(A)) = A \text{ as } \varphi \text{ is onto}$$

Cor: If $x \in A$ & $x \neq 1_A$ then $\exists m \subsetneq A$
 s.t. $x \in m$ and m is maximal ideal of A .

Let $I = \{x \in A : x \text{ is a non-unit}\}$

Suppose $I \leq A \Rightarrow I$ is a maximal ideal.

and any $x \in I$ where I is the only maximal ideal containing x and in fact I is the only maximal ideal possible as proper ideal must contain all non-units.

Prop: If the set of all units form an ideal of A , then I is the unique maximal ideal in A .

Defⁿ: A ring A is said to be a local ring if it has only one maximal ideal.

Example: $\mathbb{Q}$ Fields.

② $\mathbb{Z}_p$ where p is a prime.

Let $a, b \in \mathbb{Z}$, then if $a \notin \mathfrak{f}$ & $b \notin \mathfrak{f}$. then

$$\mathfrak{f} + (a) \notin \Sigma \text{ and } \mathfrak{f} + (b) \notin \Sigma$$

$\Rightarrow \exists n \& m$ with

$$x^n \in \mathfrak{f} + (a) \text{ \& } x^m \in \mathfrak{f} + (b)$$

$$\Rightarrow x^n = z_1 + r_1 a \text{ \& } x^m = z_2 + r_2 b$$

$$x^{n+m} = z_1 z_2 + z_1 r_2 b + z_2 r_1 a + r_1 r_2 ab \in \mathfrak{f}.$$

$\Rightarrow \mathfrak{f} \notin \Sigma$ (contradiction!!)

$\Rightarrow \mathfrak{f}$ is a prime ideal & $x^n \notin \mathfrak{f} \forall n \in \mathbb{N}$

$$\Rightarrow a \notin \mathfrak{f}.$$

$$\Rightarrow x \notin \bigcap_{\substack{\mathfrak{p} \triangleleft A \\ \mathfrak{p} \text{ prime}}} \mathfrak{p}$$